

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457077.72469 +/- 0.00091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.058
Transit depth (Rp/Rs)^2	3.48 +/- 2.15 [%]
Orbital Inclination (inc)	85.52 +/- 2.83
Airmass coefficient 1 (a1)	3.37 +/- 2.02
Airmass coefficient 2 (a2)	-0.7 +/- 0.38
Scatter in the residuals of the lightcurve fit is	56.67 %
Best Comparison Star	#2 - [36, 335]
Optimal Aperture	2.41
Optimal Annulus	6.25
Transit Duration (day)	0.0589 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.058 vs 0.1594 (deviation 0.42σ)
Duration fit vs published	0.0589 d vs 0.0512 d (deviation 1.16σ)
Mid-time O–C	1.9 min
Depth SNR	3.0
Residual scatter	56.67 %

Light curve

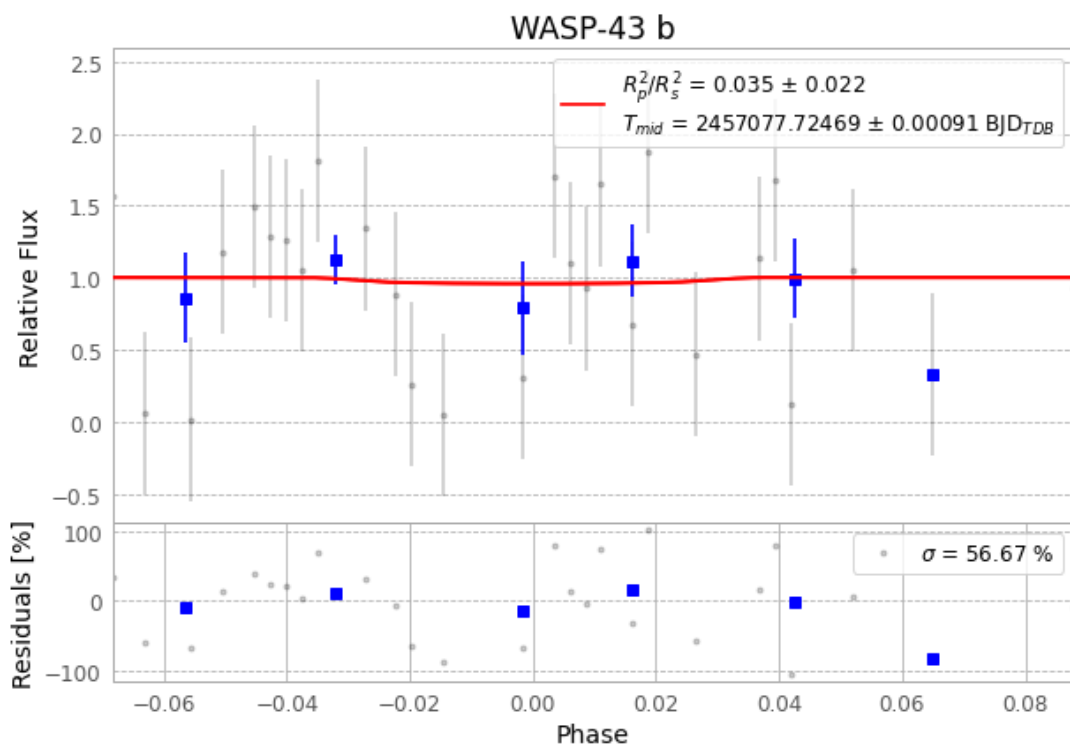


Figure 1 — FinalLightCurve_WASP-43 b_2015-02-23.png (SHA-256 9764a56d430bb2ee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 153], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a7ba341374b6e08d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

63 FITS frames (41 MB), WASP-43150224035112.FITS → WASP-43150224065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150224.log (c7604517957b...), FinalLightCurve_WASP-43 b_2015-02-23.png (9764a56d430b...), FinalLightCurve_WASP-43 b_2015-02-23.csv (77eca4027acd...)

Compute resources

Wall time 132.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:14Z.